INCH-POUND

The documentation and process conversion measures necessary to comply with this document shall be completed by 10 July 2026.

MIL-PRF-19500/127X
w/AMENDMENT 4
10 March 2026
SUPERSEDING
MIL-PRF-19500/127X
w/AMENDMENT 3
1 June 2020

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICES, DIODE, SILICON, VOLTAGE REGULATOR, TYPES 1N4370 THROUGH 1N4372 AND 1N746 THROUGH 1N759 5, 2, AND 1 PERCENT VOLTAGE TOLERANCE, ENCAPSULATED (AXIAL LEAD THROUGH-HOLE AND SURFACE MOUNT PACKAGES) AND UN-ENCAPSULATED (DIE), QUALITY LEVELS JAN, JANTX, JANTXV, JANHC, AND JANKC

JANS level (see 6.6).

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 500 milliwatt, silicon, voltage regulator diodes with voltage tolerances of 5, 2, and 1 percent. Three levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500. Two levels of product assurance is provided for each unencapsulated device.

1.2 Physical dimensions. The device package for the encapsulated device type are as follows: Axial leaded (DO-35) in accordance with figure 1, glass body surface mount (DO-213AA) in accordance with figure 2, UB, in accordance with figure 3, and UB2 in accordance with figure 4. The dimensions and topography for JANHC and JANKC unencapsulated die are as follows: A version die in accordance with figure 5, B version die in accordance with figure 6, C version die in accordance with figure 7, D version die in accordance with figure 8, and E version die in accordance with figure 9.

1.3 Maximum ratings. Maximum ratings are as shown in maximum and primary test ratings (see 3.6.1 herein), note 1, and as follows:

a. P_{TL} = 500 mW, (DO-35) at T_L = +50°C, L = .375 inch (9.53 mm); both ends of case or diode body to heat sink. (Derate I_Z to 0 at +175°C.)

b. P_{TEC} = 500 mW, (DO-213AA) at T_{EC} = +125°C, (Derate to 0 at +175°C.)
-65°C ≤ T_J ≤ +175°C; -65°C ≤ T_{STG} ≤ +175°C.

c. $P_{T(PCB)}$ = 400 mW, T_A = 55°C (see note 2). (Derate to 0 at +175°C.)

d. $P_{SP(IS)}$ = 500 mW (UB) at $T_{SP(IS)}$ = 125°C. (Derate to 0 at +175°C.)
-65°C ≤ T_J ≤ +175°C; -65°C ≤ T_{STG} ≤ +175°C.

Comments, suggestions, or questions on this document should be addressed to DLA Weapons Support, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.3 Maximum ratings. – continued.

(1) See figures 10, 11, and 12 for derating curves.

(2) T_A = +75°C for both axial leaded and UR mounted horizontally on printed circuit board (PCB) and in still air. The details for the PCB and in still air are as follows. Base material is FR4 .0625 inch (1.59 mm) thick, 1-layer 1-oz Cu. PCB pad size for UR devices = .067 inch (1.70 mm) x .105 inch (2.67 mm); PCB land sizes for axial leaded devices is .092 inch (2.34 mm) diameter with L ≤ .187 inch (4.75 mm) lead length. Strips connected to the lands and pads are .030 inch (0.762 mm) x 1 inch (25.4 mm) long.

1.4 Primary electrical characteristics. Primary electrical characteristics are as shown in maximum and primary test ratings (see 3.6.1 herein) and as follows:

a. 2.4 V dc ≤ V_Z ≤ 12 V dc.

b. 1N4370A through 1N4372A and 1N746A through 1N759A are ±5 percent voltage tolerance.

c. 1N4370C through 1N4372C and 1N746C through 1N759C-1 are ±2 percent voltage tolerance.

d. 1N4370D through 1N4372D and 1N746D through 1N759D-1 are ±1 percent voltage tolerance.

e. For thermal impedance curves, see figures 13, 14, 15, 16, and 17.

f. $R_{\theta JEC}$ = 100°C/W maximum. Junction to end-caps (DO-213AA).

g. $R_{\theta JA}$ = 300°C/W junction to ambient including PCB. $R_{\theta JA}$ with a defined thermal resistance condition included is measured at I_Z = as defined in the electrical characteristics table herein.

h. $R_{\theta JSP(IS)}$ = 90°C/W (maximum) junction to solder pad (IS) (UB).

i. $R_{\theta JA(PCB)}$ = 250°C/W (maximum) junction to ambient (PCB) (UB).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level designators.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", and "JANTXV".

1.5.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 Device type. The designation system for the device types of semiconductor covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The semiconductors of this specification sheet use the first number and letter symbols "1N".

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1.5.2.2 Second number symbols. The second number symbols for the semiconductor covered by this specification sheet are as follows: "1N4370", "1N4371", "1N4372", "1N746", "1N747", "1N748", "1N749", "1N750", "1N751", "1N752", "1N753", "1N754", "1N755", "1N756", "1N757", "1N758", and "1N759".

1.5.3 Suffix symbols. The following suffix symbol(s) are incorporated in the PIN as applicable.

1.5.3.1 First suffix symbol (voltage tolerance). The first suffix letter "A" indicates that the semiconductor is a ±5 percent voltage tolerance. The first suffix letter "C" indicates that the semiconductor is a ±2 percent voltage tolerance. The first suffix letter "D" indicates that the semiconductor is a ±1 percent voltage tolerance device type.

1.5.3.2 Second suffix symbols (package designators). The following suffix symbols that designate the package outline for the devices covered by this specification sheet are as follows.

-1	Indicates an axial through hole DO-35 metallurgical bonded double plug construction (see figure 1).
UR-1	Indicates a surface mount DO-213AA using a metallurgical bonded double plug construction (see figure 2)
UB	Indicates a 3 terminal, 4 pad surface mount package. The metal lid is connected to pad 4 (see figure 3).
UB2	Indicates a 2 terminal, 4 pad surface mount package. The metal lid is connected to pad 4 (see figure 4).

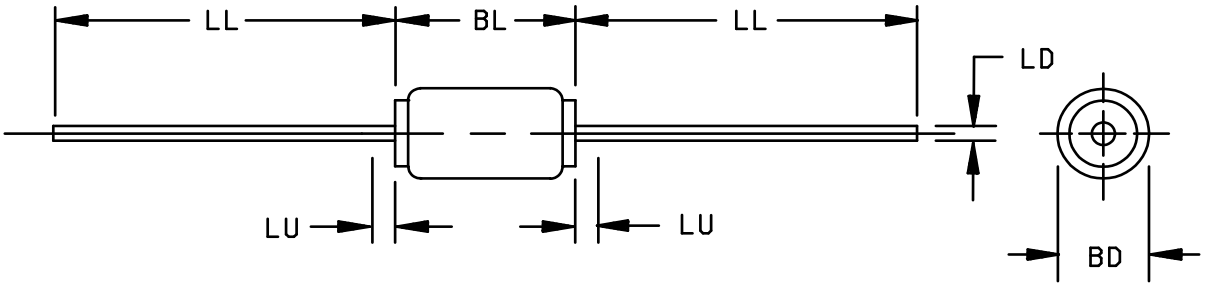
1.5.3.3 Internal circuit schematic designator suffix synbols (for UB and UB2 packages only). The suffix symbols that designate the internal circuit schematic (see schematics on figure 3) of the surface mount packages are as follows:

	Indicates a unidirectional, single diode circuit (UB package only).
CA	Indicates dual diode with a common anode pin (UB package only).
CC	Indicates a dual diode with a common cathode pin (UB package only).
D	Indicates a dual diode doubler (UB package only).
R	Indicates a reverse polarity configuration (UB2 package only).

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

1.5.5 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A", "B", "C", "D", and "E". (see figures 5 through 9 and 6.5.2)

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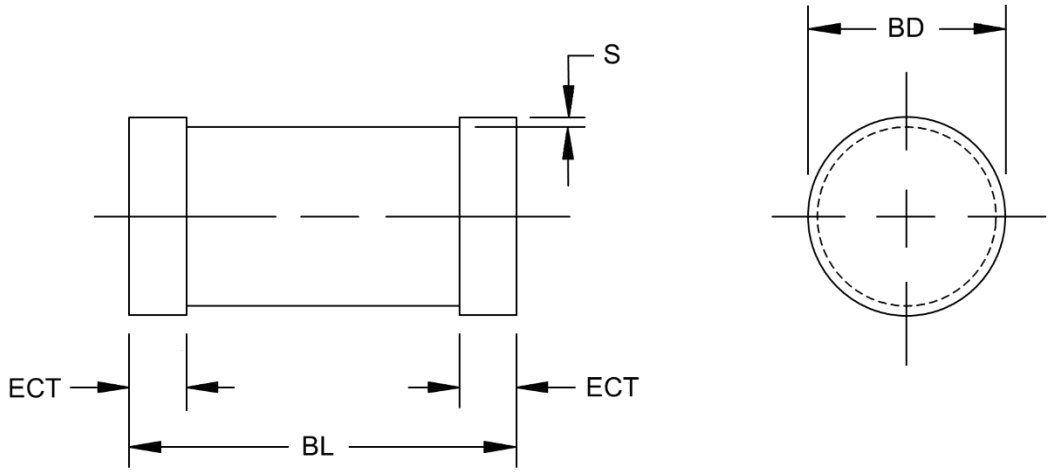


Symbol	Dimensions				Notes
	inch	inch	mm	mm	
	Min	Max	Min	Max	
BD	.055	.090	1.40	2.29	2
BL	.120	.200	3.05	5.08	3 4
LD	.018	.023	0.46	0.58	4
LL	1.000	1.500	25.40	38.10	
LU		.050		1.27	5

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. Dimension BD shall be measured at the largest diameter.
4. The BL dimension shall include the entire body including slugs.
5. Dimension LU shall include the sections of the lead over which the diameter is uncontrolled. Lead diameter not controlled in this zone to allow for flash, lead finish build-up and minor irregularities other than slugs. This uncontrolled area is defined as the zone between the edge of the diode body and extending .050 inch (1.27 mm) onto the leads.

FIGURE 1. Physical dimensions for axial leaded DO-204AH package.

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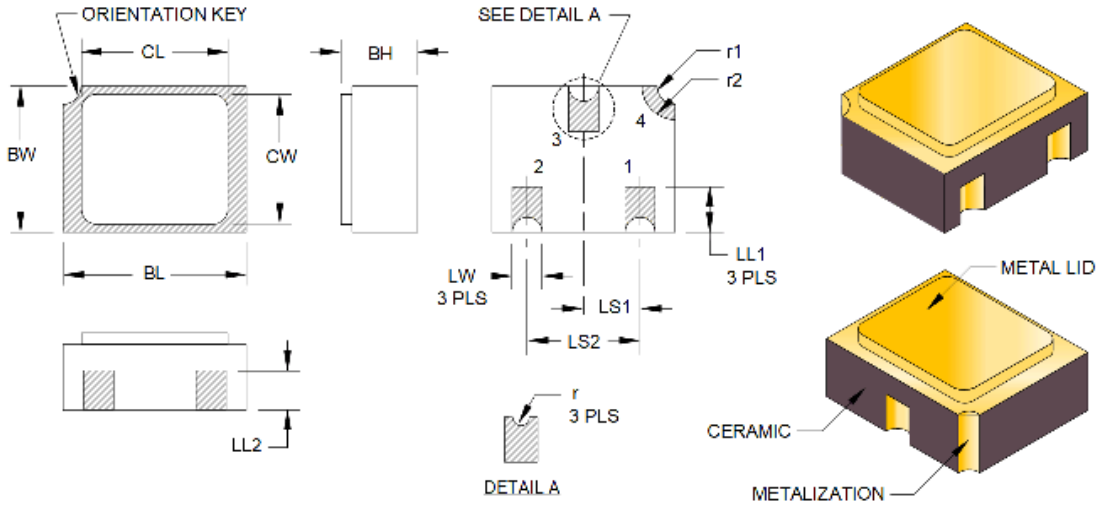


Symbol	Dimensions				
	inch	inch	mm	mm	Note
	Min	Max	Min	Max	
BL	.130	.146	3.30	3.71	
BD	.063	.067	1.60	1.71	2, 3
ECT	.016	.022	0.41	0.56	
S	0		0		3

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Dimensions are pre-solder dip.
3. Dimension S is optional however, dimension BD (glass body) shall not exceed endcap diameter.

FIGURE 2. Physical dimensions for DO-213AA (UR) surface mount package.

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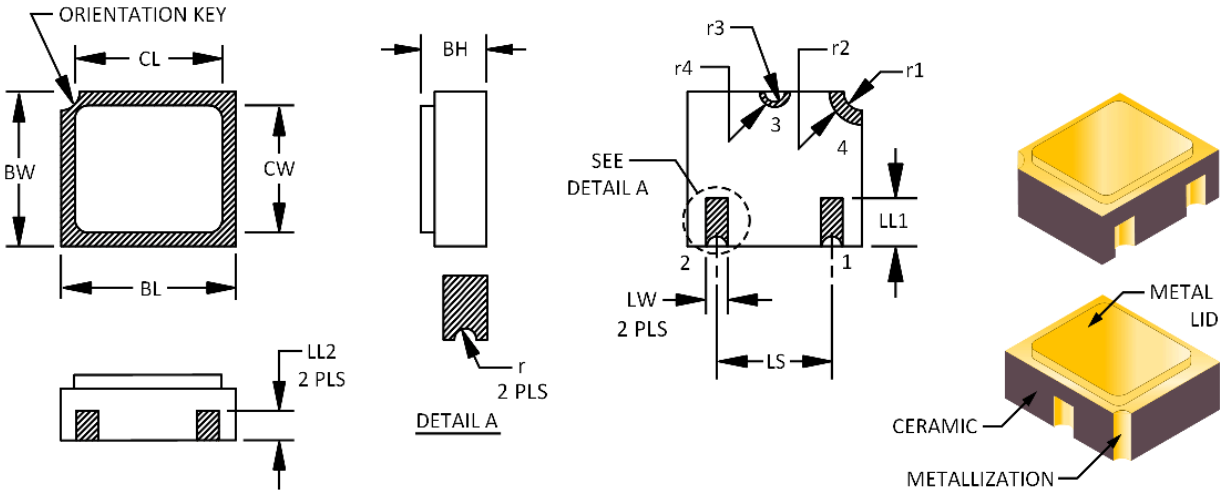


Symbol	Dimensions				Symbol	Dimensions			
	inch	inch	mm	mm		inch	inch	mm	mm
	Min	Max	Min	Max		Min	Max	Min	Max
BH	.040	.056	1.02	1.42	LS1	.035	.039	0.89	0.99
BL	.115	.128	2.92	3.25	LS2	.071	.079	1.80	2.01
BW	.085	.108	2.16	2.74	LW	.016	.024	0.41	0.61
CL		.128		3.25	r		.008		0.20
CW		.108		2.74	r1		.012		0.31
LL1	.022	.038	0.56	0.97	r2		.022		0.56
LL2	.017	.035	0.43	0.89					

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Ceramic package only.
3. Hatched areas on package denote metallized areas. Pad 4 = shielding, connected to the lid.
4. Dimensions are pre-solder dip.

FIGURE 3. Physical dimensions, 3 terminal UB surface mount package.

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POLARITY

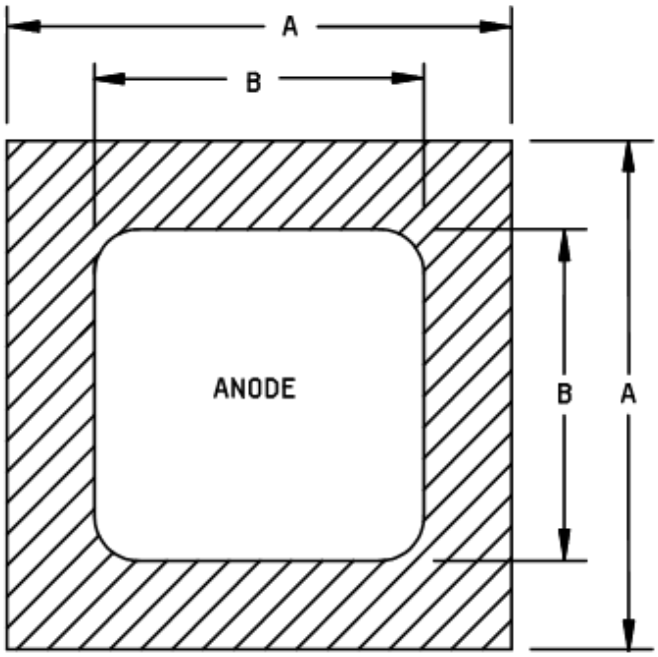


Symbol	Dimensions				Symbol	Dimensions			
	inch	inch	mm	mm		inch	inch	mm	mm
	Min	Max	Min	Max		Min	Max	Min	Max
BH	.046	.056	1.17	1.42	LS	.071	.079	1.80	2.01
BL	.115	.128	2.92	3.25	LW	.016	.024	0.41	0.61
BW	.085	.108	2.16	2.74	r	.008 TYP		0.20 TYP	
CL		.128		3.25	r1	.012 TYP		0.30 TYP	
CW		.108		2.74	r2	.022 TYP		0.56 TYP	
LL1	.022	.038	0.56	0.96	r3	.008 TYP		0.20 TYP	
LL2	.017	.035	0.43	0.89	r4	.012 TYP		0.30 TYP	

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Ceramic package only.
3. Hatched areas on package denote metallized areas. Pad 4 = shielding, connected to the lid.
4. Dimensions are pre-solder dip.

FIGURE 4. Physical dimensions, 2 terminal UB2 surface mount package.

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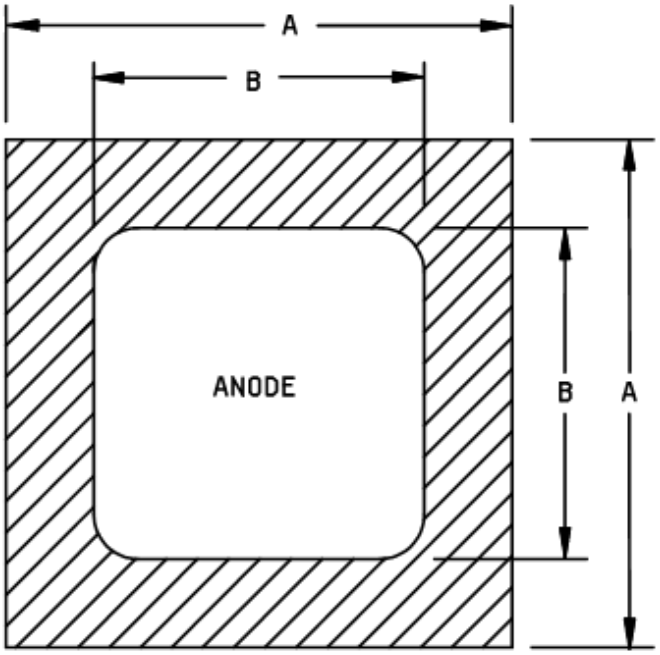
(A - version)

Ltr	Dimensions			
	inch	inch	mm	mm
	Min	Max	Min	Max
A	.021	.025	0.53	0.63
B	.013	.017	0.33	0.43

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. The die thickness is .010 ±.002 (0.25 mm ±0.051 mm).
3. The physical characteristics of the die are as follows:
   Metallization is top (anode) = AL, back (cathode) = AU.
   AL thickness = 25,000 Å minimum,
   AU thickness = 4,000 Å minimum.

FIGURE 5. Physical dimensions for JANHCA and JANKCA die.

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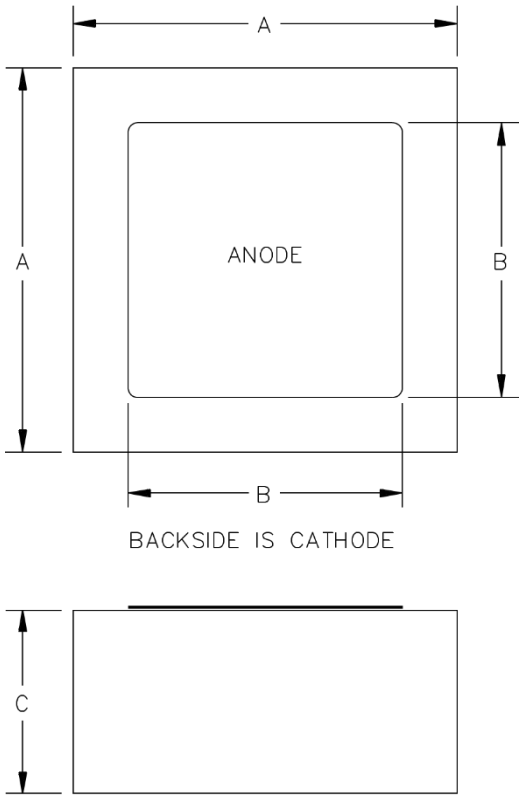
(B – version)

Ltr	Dimensions			
	inch	inch	mm	mm
	Min	Max	Min	Max
A	.024	.028	0.61	0.71
B	.017	.021	0.43	0.53

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. The die thickness is .010 ±.002 (0.25 mm ±0.051 mm).
3. Circuit layout data: For zener operation, cathode must be operated positive with respect to anode.
4. The physical characteristics of the die are as follows:
   Metallization is top (anode) = Al, back (cathode) = Au.
   Al thickness = 40,000 Å minimum,
   Au thickness = 5,000 Å minimum.

FIGURE 6. Physical dimensions for JANHCB and JANKCB die.

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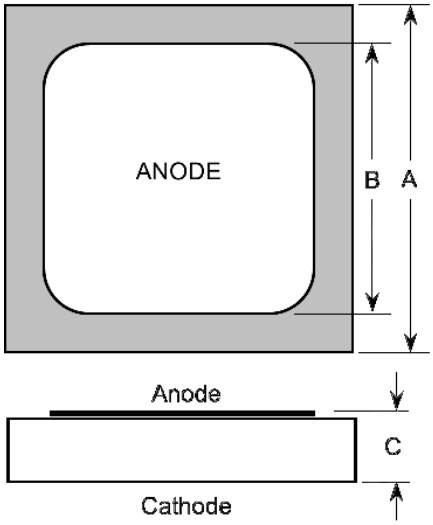


Ltr	Dimensions			
	inch	inch	mm	mm
	Min	Max	Min	Max
A	.019	.023	0.48	0.58
B	.013	.017	0.33	0.43
C	.008	.012	0.20	0.30

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Element evaluation accomplished utilizing TO-5 package.
3. The physical characteristics of the die are as follows:
   Metallization is top (anode) = Al, back (cathode) = Au
   Al thickness = 25,000 Å minimum.
   Au thickness = 4,000 Å minimum.

FIGURE 7. Physical dimensions for JANHCC and JANKCC die.

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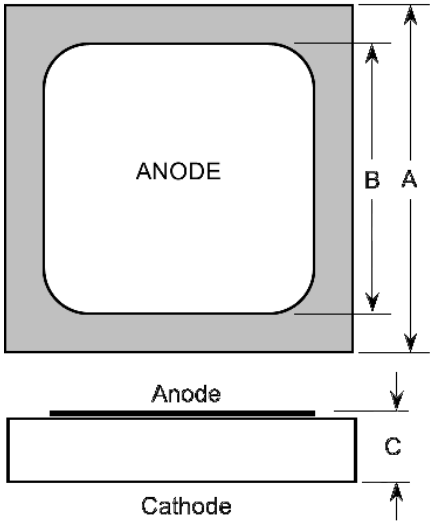


Ltr	Dimensions			
	inch	inch	mm	mm
	Min	Max	Min	Max
A	.023	.027	0.584	0.686
B	.015	.019	0.381	0.483
C	.008	.012	0.203	0.305

NOTES:
1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. The physical characteristics of the die are as follows:
   Metallization is top (anode) = Al, back (cathode) = Ti / Au
   Al thickness = 34,000 Å minimum;
   Tl / Au thickness = 200 Å / 4,350 Å nominal.

FIGURE 8. Physical dimensions for JANHCD and JANKCD die.

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Ltr	Dimensions			
	Inch	Inch	mm	mm
	Min	Max	Min	Max
A	.011	.015	0.28	0.38
B	.005	.009	0.13	0.23
C	.004	.008	0.10	0.20

NOTES:
1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. The physical characteristics of the die are as follows:
   Metallization is top (anode) = AL, back (cathode) = AU
   Al thickness = 20,000 Å minimum;
   Au thickness = 8,000 Å nominal.

FIGURE 9. Physical dimensions for JANHCE and JANKCE die.

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2. APPLICABLE PUBLICATIONS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML-19500) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be specified in MIL-PRF-19500 and figure 1 and figure 2 (DO-35 and DO-213AA), figure 3 (UB) and figure 4 (UB2) and figures 5, 6, 7, 8, and 9 (die), herein.

3.4.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Diode construction.

3.4.2.1 Axial leaded and round body surface mount. All devices shall be metallurgically bonded double plug construction in accordance with the requirements of category I, II, or III (see MIL-PRF-19500). UR version devices shall be structurally identical to the non-surface mount devices except for lead terminations.

3.4.2.2 Bottom solder pad surface mount. The UB and UB2 version devices die shall be eutectically mounted and wire bonded in a ceramic package.

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3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. Manufacturer's identification and date code shall be marked on the devices. Initial container package marking shall be in accordance with MIL-PRF-19500. The prefixes JAN, JANTX, or JANTXV can be abbreviated as J, JX, or JV, respectively. (Example: The PIN for an axial leaded device may be reduced from JANTX1N4370D−1 to JX4370D−1). All marking which is omitted from the body of the device, except for polarity and serial numbers, shall appear on the initial container.

3.5.1 Axial leaded devices. For axial leaded versions only, the manufacturer's identification and date code shall be marked on the devices. The polarity shall be indicated with a contrasting color band to denote the cathode end. No color coding shall be permitted for part identification. No color coding shall be permitted for part identification.

3.5.2 UR devices. For 'UR' version devices only, all marking, except polarity may be omitted from the body, but shall be retained on the initial container. Polarity marking of 'UR' devices shall consist as a minimum, a band or 3 contrasting dots around the periphery of the cathode. At the option of the manufacturer, 'UR' devices may include laser marking on an end-cap, to include PIN and lot date code for all levels.

3.5.3 Bottom solder pad surface mount devices. The 'UB' and 'UB2' devices do not require polarity marking.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, table I, table II, and table III.

3.6.1 Maximum and primary test ratings. Maximum and primary test ratings for voltage regulator diodes are specified in table IV herein.

3.6.2 Selection of tight tolerance devices. The C and D suffix devices shall be selected from JAN, JANTX, or JANTXV devices, which have successfully completed all applicable screening, and groups A, B, and C testing as 5 percent tolerance devices. All sublots of C and D suffix devices shall pass table I, subgroup 2, at tightened tolerances. Tighter tolerances for mounting clip temperature shall be maintained for reference purpose to establish correlation. For C and D tolerance levels, T_L = +25°C ±2°C at .375 inch (9.53 mm) from body, or zero inches for surface mount devices or equivalent.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

d. Element evaluation of un-encapsulated die (see 4.6).

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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.2.2 JANHC and JANKC devices. JANHC and JANKC devices shall be qualified in accordance with appendix G of MIL-PRF-19500.

4.2.3 Construction verification. Cross sectional photos from three devices shall be submitted in the qualification report.

4.3 Screening (JAN, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for JANTX and JANTXV levels
3a	Temperature cycling
(1) 3c	Thermal impedance (see 4.3.2)
7a 7b	Not applicable Optional
9	Not applicable
11	I_{R1} and V_{Z2}
12	See 4.3.3
(2) 13	ΔI_{R1} ≤ 100 percent of initial reading or 50 nA dc, whichever is greater. ΔV_{Z2} ≤ ±2 percent initial reading. Subgroup 2 of table I herein.
14a (3) 14b	Not applicable Required

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.
(2) PDA = 5 percent for screen 13 applies to ΔI_{R1} and ΔV_{Z2}, and I_{R1} and V_{Z2}. Thermal impedance ($Z_{\theta JX}$) is not required in screen 13 of 4.3 herein.
(3) For clear glass diodes, the hermetic seal (gross leak) may be performed at anytime after temperature cycling.

4.3.1 JAN product. JAN product will have temperature cycling and thermal impedance testing performed in accordance with MIL-PRF-19500, JANTX level screening requirements.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 of MIL-STD-750, as applicable, using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). (See figures 13, 14, 15, 16, and 17). See table II, subgroup 4 herein.

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4.3.3 Power burn-in conditions. Power burn-in conditions are as follows (see 4.5.5): $I_{Z(min)}$ = column 8 of table IV herein. T_A = 75°C maximum. Test conditions in accordance with method 1038 of MIL-STD-750, condition B. Adjust I_Z or T_A to achieve the required T_J. T_J = 125°C minimum. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. Group A inspection shall be performed on each sublot.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein. ($Z_{\theta JX}$ applies to B2 and B3 only).

Subgroup	Method	Condition
B2	1056	0°C to +100°C, 10 cycles.
B2	1051	-55°C to +175°C, 25 cycles.
B2	4066	Condition A, See 4.5.1.
B3	1027	$I_{Z(min)}$ = Column 8 of table IV minimum. Adjust I_{ZM} or T_A to ensure a T_J = 150°C (minimum) (see 4.5.5).
B4	2075	As applicable.
B4	2101	Decap analysis, scribe and break only.
B6	1032	T_A = +175°C.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. ($Z_{\theta JX}$ applies to C2 and C6 only).

Subgroup	Method	Condition
C2	1056	0°C to +100°C, 10 cycles.
C2	1051	-55°C to +175°C, 25 cycles.
C2	2036	Tension: Test condition A; weight = 10 pounds (4.54 kg), t = 15 seconds. Lead fatigue: Test condition E. (Tension and lead fatigue are not required for UR-1, UB or UB2 suffix devices).
C2	1071	Test condition E for axial leaded and UR devices (not applicable for UB and UB2). Fine and gross leak are required for UB and UB2.
C3		Not applicable.
C5	4081	See 4.3.2 herein.
C6	1027	$I_{Z(min)}$ = Column 8 of table IV minimum. Adjust I_{ZM} or T_A to ensure a T_J = 150°C (minimum) (see 4.5.5).
C7		Not applicable.
C8	4071	I_Z = 7.5 mA dc, T_1 = +25°C ±5°C, T_2 = +125°C ±5°C. (Max limit in accordance with columns 13 and 14 of table IV). Sample size = 22, 0 rejects allowed.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and table II herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, subgroup 2 herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Surge current (I_{ZSM}). The peak currents shown in column 10 of table IV shall be applied in the reverse direction and these shall be imposed on the current (I_Z = 20 mA dc) a total of 5 surges at 1 minute intervals. Each individual surge shall be one-half square-wave-pulse of one one-hundred twenty second duration or an equivalent one-half sine wave with the same effective rms current. T_A = +25°C ±5°C.

4.5.2 Regulator voltage measurements. The test current shall be applied until thermal equilibrium is attained (20 ±2 seconds) prior to reading the breakdown voltage. For this test, the diode shall be suspended by its leads with mounting clips whose inside edge is located at .375 inch (9.53 mm) from the body (UR version = 0 lead length) and the mounting clips shall be maintained at a temperature of +25°C +8°C, and -2°C. This measurement may be performed after a shorter time, following application of the test current, than that which provides thermal equilibrium if correlation to stabilized readings can be established to the satisfaction of the qualifying activity. JANHC and JANKC shall be pulse tested at 10 ms maximum.

4.5.3 Voltage regulation $V_{Z(reg)}$. Voltage regulation shall be determined by the difference of the regulator voltage measured at different currents as specified in table I, subgroup 7. Both tests shall be performed at thermal equilibrium. This ΔV_Z shall not exceed column 7 of table IV.

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4.5.4 Temperature coefficient of regulator voltage (α_{VZ}). The device shall be temperature stabilized with current applied prior to reading regulator voltage at the specified ambient temperature as specified in 4.4.3 herein, group C, subgroup 8.

4.5.5 Free air burn-in and life tests. The use of a current limiting or ballast resistor is permitted provided that each DUT still sees at least the $I_{Z(min)}$ described in 4.3.2 and that the minimum applied voltage, where applicable, is maintained throughout the burn-in period. Use method 3100 of MIL-STD-750 to measure T_J.

4.6 Element evaluation of unencapsulated die. The element evaluation of unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500. The test of subgroups 3 and higher shall be performed in any convenient package. For subgroup 4, the burn-in duration for the JANKC level shall follow the JANS requirements and the JANHC shall follow the JANTX requirements.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit 2/	Limit 2/	Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Forward voltage	4011	Condition A, I_F = 200 mA dc	V_F		1.1	V dc
Reverse current	4016	Condition A or B, DC method, V_R = column 11 of table IV	I_{R1}		Col. 12	µA dc
Regulator voltage	4022	I_{Z1} = 250 µA dc (see 4.5.2)	V_{Z1}	Col. 9	Col. 4	V dc
Regulator voltage	4022	I_{Z2} = 20 mA dc (see 4.5.2)	V_{Z2}	Col. 3	Col. 4	V dc
Thermal impedance 3/	3101	See 4.3.2	$Z_{\theta JX}$			°C/W
Subgroup 3						
High temperature operation		T_A = 150°C				
Reverse current	4016	Condition A or B, DC method, V_R = column 11 of table IV	I_{R2}		Col. 5	µA dc
Subgroup 4						
Small-signal reverse breakdown impedance	4051	I_Z = 20 mA dc, I_{SIG} = 10 percent of I_Z ac	Z_Z		Col. 6	ohm
Subgroups 5						
Not applicable						
Subgroup 6						
Surge current	4066	Condition A, See 4.5.1				
Electrical measurements		See table I, subgroup 2				
Subgroup 7						
Voltage regulation		I_{Z3} = 2 mA dc; I_{Z4} = 20 mA dc (see 4.5.3)	$V_{Z(reg)}$		Col. 7	V dc

1/ For sampling plan, see MIL-PRF-19500.
2/ Column references are to table IV.
3/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroups 3, 4 and 5 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, JANTXV).
Group C, subgroups 2 and 6.
Group E, subgroup 1.

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TABLE II. Group E inspection qualification and requalification (all product assurance levels).

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Temperature cycling	1051	500 cycles, -55°C to +175°C	
Hermetic seal	1071		
Electrical measurements		See table I, subgroup 2 and table III, step 1	
Subgroup 2			22 devices, c = 0
Intermittent operation life	1037	6,000 cycles. I_Z = column 8 of table IV	
Electrical measurements		See table I, subgroup 2 and table III, step 1. Provide read and record $\Delta Z_{\theta JX}$ data to the qualifying activity.	
Subgroup 4			N/A
Thermal impedance curves		See MIL-PRF-19500	
Subgroups 5 and 6			
Not applicable			
Subgroup 8			45 devices
Resistance to glass cracking	1057	Condition B. Cool down after solder immersion is permitted. Test until failure occurs on all devices with the chosen sample or to a maximum of 25 cycles, whichever comes first.	

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TABLE III. Group B, C, and E delta measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limit	Limit	Unit
		Method	Conditions		Min	Max	
1.	Thermal impedance	3101	See 4.3.2	$\Delta Z_{\theta JX}$		±10 percent of initial value	°C/W

1/ The electrical measurements for table E-VIb of MIL-PRF-19500 are as follows:
Subgroup 2, see table III herein, step 1.
Subgroup 3, see table III herein, step 1.
2/ The electrical measurements for table E-VII of MIL-PRF-19500 are as follows:
Subgroup 2, see table III herein, step 1.
Subgroup 6, see table III herein, step 1.
3/ The electrical measurements for table E-IX of MIL-PRF-19500 are as follows:
Subgroup 1, see table III herein, step 1.
Subgroup 2, see table III herein, step 1.

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TABLE IV. Electrical characteristics.

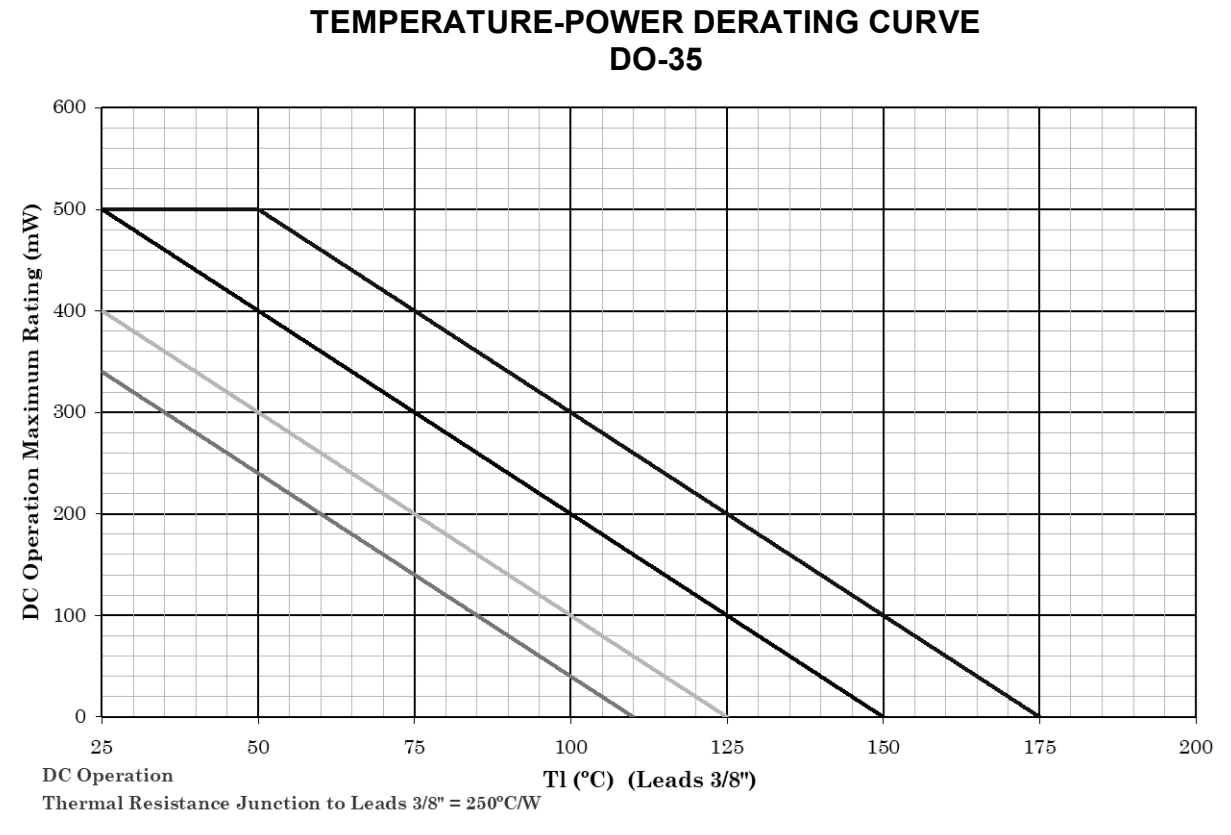
Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10		Col 11	Col 12	Col 13	Col 14
Type	V_{Z2} nom at I_Z = 20 mA	V_{Z2} Min	V_{Z1} and V_{Z2} max	I_{R2} at T_A = 150°C V_R = col 11	Z_Z at I_Z =20 mA	$V_{Z(reg)}$	I_{ZM}	$V_{Z1(min)}$ Z_{ZK}	I_{ZSM}	I_{ZSM} (surge) UB, UB2	V_R	I_{R1} at T_A = 25°C V_R = col 11	α_{VZ}	α_{VZ} pos. limit
	Volts	Volts	Volts	µA dc	Ohms	Volts	mA dc	Volts	mA dc	mA dc	Volts	µA dc	%/°C	%/°C
1N4370A	2.4	2.28	2.52	200	30	1.0	155	1.1	1000	313	1.0	100	-0.085	0
1N4371A	2.7	2.57	2.83	150	30	1.0	140	1.2	1000	313	1.0	60	-0.080	0
1N4372A	3.0	2.85	3.15	100	29	1.0	125	1.3	1000	313	1.0	30	-0.075	0
1N746A	3.3	3.14	3.46	30	24	1.0	120	1.5	1000	313	1.0	5	-0.070	0
1N747A	3.6	3.42	3.78	30	22	1.0	110	1.8	1000	313	1.0	3	-0.065	0
1N748A	3.9	3.71	4.09	30	20	1.0	100	2.0	1000	313	1.0	2	-0.060	0
1N749A	4.3	4.09	4.51	50	18	1.0	90	2.4	990	309	1.0	2	-0.055	+.020
1N750A	4.7	4.47	4.93	50	15	1.0	85	2.8	980	306	1.5	5	-0.043	+.025
1N751A	5.1	4.85	5.35	50	14	0.8	75	3.3	960	300	2.0	5	-0.030	+.030
1N752A	5.6	5.32	5.88	50	8	0.8	70	4.3	950	297	2.5	5	-0.028	+.036
1N753A	6.2	5.89	6.51	50	3	0.6	65	5.2	910	284	3.5	5	0	+.045
1N754A	6.8	6.46	7.14	50	3	0.4	60	6.0	870	272	4.0	2	0	+.050
1N755A	7.5	7.13	7.87	50	4	0.4	55	6.6	810	253	5.0	2	0	+.058
1N756A	8.2	7.79	8.61	50	5	0.4	50	7.5	740	231	6.0	1	0	+.062
1N757A	9.1	8.65	9.55	50	6	0.5	45	8.4	650	203	7.0	1	0	+.068
1N758A	10.0	9.50	10.50	50	7	0.7	40	9.1	540	169	8.0	1	0	+.076
1N759A	12.0	11.40	12.60	50	10	1.0	35	11.0	400	125	9.0	1	0	+.080
1N4370C	2.4	2.352	2.448	200	30	1.0	155	1.1	1000	313	1.0	100	-0.085	0
1N4371C	2.7	2.646	2.754	150	30	1.0	140	1.2	1000	313	1.0	60	-0.080	0
1N4372C	3.0	2.94	3.06	100	29	1.0	125	1.3	1000	313	1.0	30	-0.075	0
1N746C	3.3	3.234	3.366	30	24	1.0	120	1.5	1000	313	1.0	5	-0.070	0
1N747C	3.6	3.528	3.672	30	22	1.0	110	1.8	1000	313	1.0	3	-0.065	0
1N748C	3.9	3.822	3.978	30	20	1.0	100	2.0	1000	313	1.0	2	-0.060	0
1N749C	4.3	4.214	4.386	50	18	1.0	90	2.4	990	309	1.0	2	-0.055	+.020
1N750C	4.7	4.606	4.794	50	15	1.0	85	2.8	980	306	1.5	5	-0.043	+.025
1N751C	5.1	4.998	5.202	50	14	0.8	75	3.3	960	300	2.0	5	-0.030	+.030
1N752C1	5.6	5.488	5.712	50	8	0.8	70	4.3	950	297	2.5	5	-0.028	+.036

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TABLE IV. Electrical characteristics - Continued.

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10		Col 11	Col 12	Col 13	Col 14
Type	V_{Z2} nom at I_Z = 20 mA	V_{Z2} min	V_{Z1} and V_{Z2} max	I_{R2} at T_A = 150°C V_R = col 11	Z_Z at I_Z = 20 mA	$V_{Z(reg)}$	I_{ZM}	$V_{Z1(min)}$ Z_{ZK}	I_{ZSM}	I_{ZSM} (surge) UB, UB2	V_R	I_{R1} at T_A = 25°C V_R = col 11	αVZ	αVZ pos. limit
	Volts	Volts	Volts	µA dc	Ohms	Volts	mA dc	Volts	mA dc	mA dc	Volts	µA dc	%/°C	%/°C
1N753C	6.2	6.076	6.324	50	3	0.6	65	5.2	910	284	3.5	5	0	+.045
1N754C	6.8	6.664	6.936	50	3	0.4	60	6.0	870	272	4.0	2	0	+.050
1N755C	7.5	7.357	7.650	50	4	0.4	55	6.6	810	253	5.0	2	0	+.058
1N756C	8.2	8.036	8.364	50	5	0.4	50	7.5	740	231	6.0	1	0	+.062
1N757C	9.1	8.918	9.282	50	6	0.5	45	8.4	650	203	7.0	1	0	+.068
1N758C	10.0	9.80	10.20	50	7	0.7	40	9.1	540	169	8.0	1	0	+.076
1N759C	12.0	11.76	12.24	50	10	1.0	35	11.0	400	125	9.0	1	0	+.080
1N4370D	2.4	2.376	2.424	200	30	1.0	155	1.1	1000	313	1.0	100	-0.085	0
1N4371D	2.7	2.673	2.727	150	30	1.0	140	1.2	1000	313	1.0	60	-0.080	0
1N4372D	3.0	2.970	3.030	100	29	1.0	125	1.3	1000	313	1.0	30	-0.075	0
1N746D	3.3	3.267	3.333	30	24	1.0	120	1.5	1000	313	1.0	5	-0.070	0
1N747D	3.6	3.564	3.636	30	22	1.0	110	1.8	1000	313	1.0	3	-0.065	0
1N748D	3.9	3.861	3.939	30	20	1.0	100	2.0	1000	313	1.0	2	-0.060	0
1N749D	4.3	4.257	4.343	50	18	1.0	90	2.4	990	309	1.0	2	-0.055	+.020
1N750D	4.7	4.653	4.747	50	15	1.0	85	2.8	980	306	1.5	5	-0.043	+.025
1N751D	5.1	5.049	5.151	50	14	0.8	75	3.3	960	300	2.0	5	-0.030	+.030
1N752D	5.6	5.544	5.656	50	8	0.8	70	4.3	950	297	2.5	5	-0.028	+.036
1N753D	6.2	6.138	6.262	50	3	0.6	65	5.2	910	284	3.5	5	0	+.045
1N754D	6.8	6.732	6.868	50	3	0.4	60	6.0	870	272	4.0	2	0	+.050
1N755D	7.5	7.425	7.575	50	4	0.4	55	6.6	810	253	5.0	2	0	+.058
1N756D	8.2	8.118	8.282	50	5	0.4	50	7.5	740	231	6.0	1	0	+.062
1N757D	9.1	9.009	9.191	50	6	0.5	45	8.4	650	203	7.0	1	0	+.068
1N758D	10.0	9.90	10.10	50	7	0.7	40	9.1	540	169	8.0	1	0	+.076
1N759D	12.0	11.88	12.12	50	10	1.0	35	11.0	400	125	9.0	1	0	+.080

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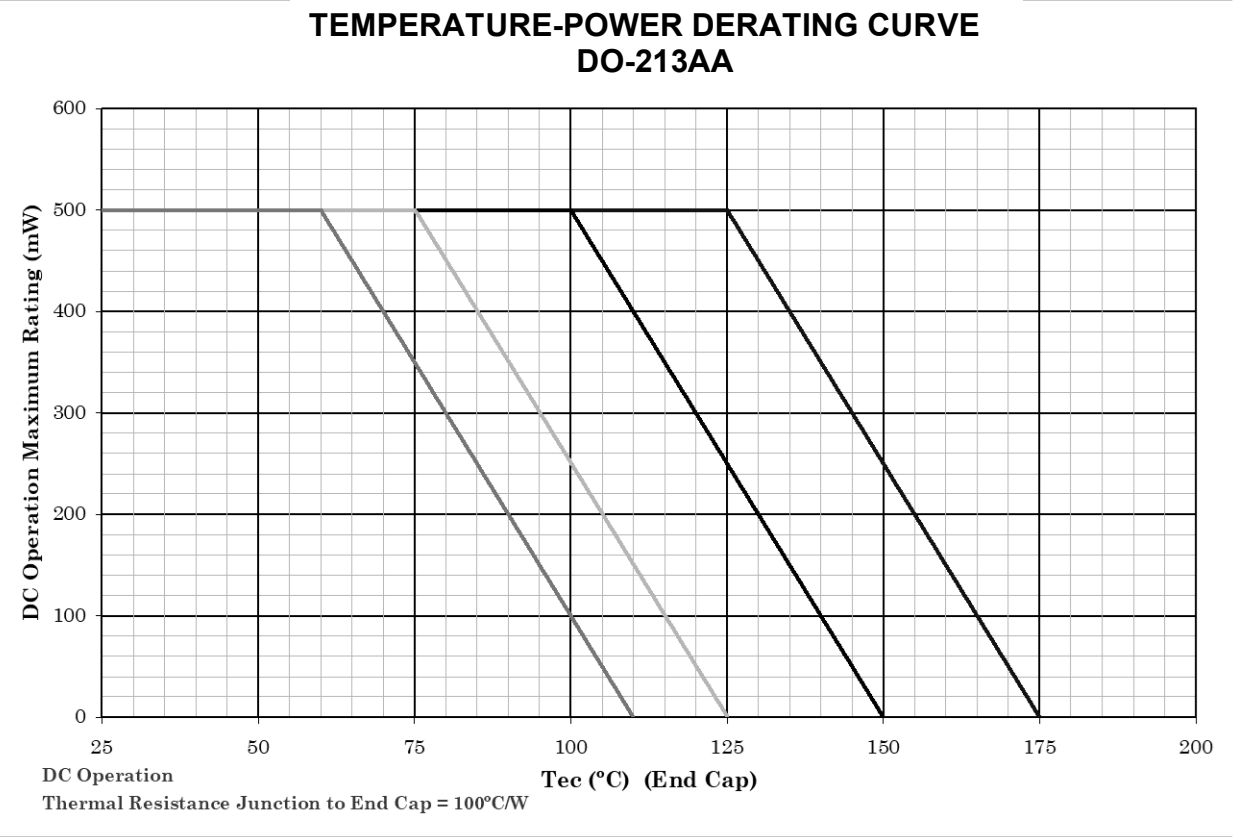


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 175°C) and power rating specified (see 1.3 herein.).
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤ 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 10. Temperature-power derating curve (junction to lead) (T_L) for axial leaded DO-35.

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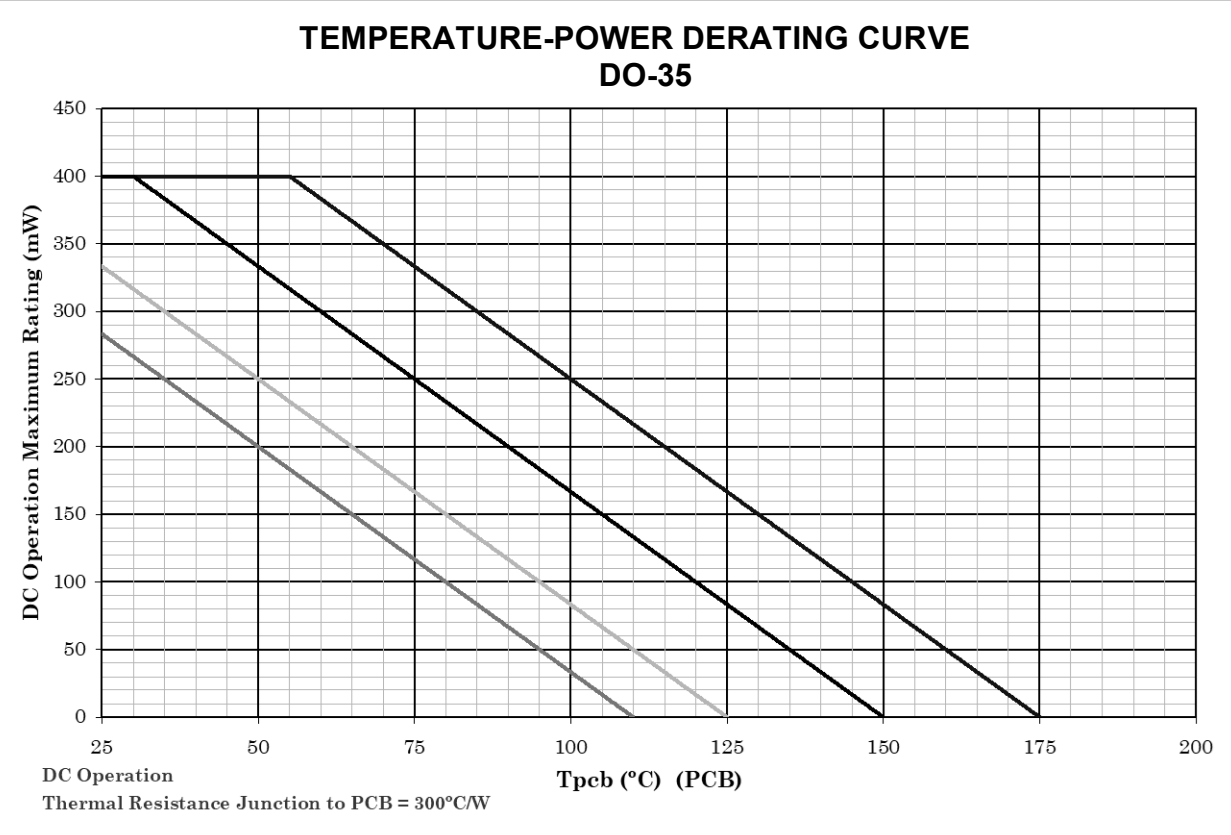


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 175°C) and power rating specified (see 1.3 herein).
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤ 125°C and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 11. Temperature-power derating curve (junction to end-cap) (T_{EC}) for UR surface mount.

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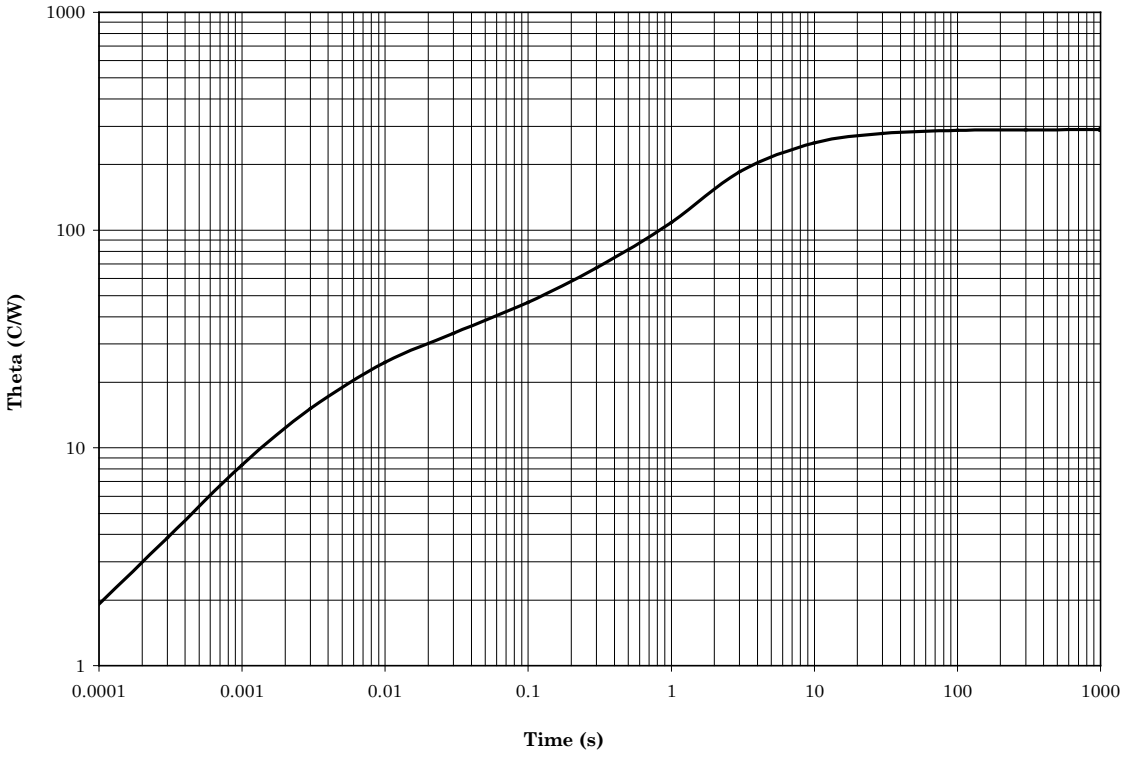
NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 175°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at T_J ≤ 125≤C, and 110°C to show power rating where most users want to limit TJ in their application.

FIGURE 12. Temperature-power derating curve (junction to PCB) (T_{PCB}) for axial leaded DO-35.

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THERMAL IMPEDANCE
PCB



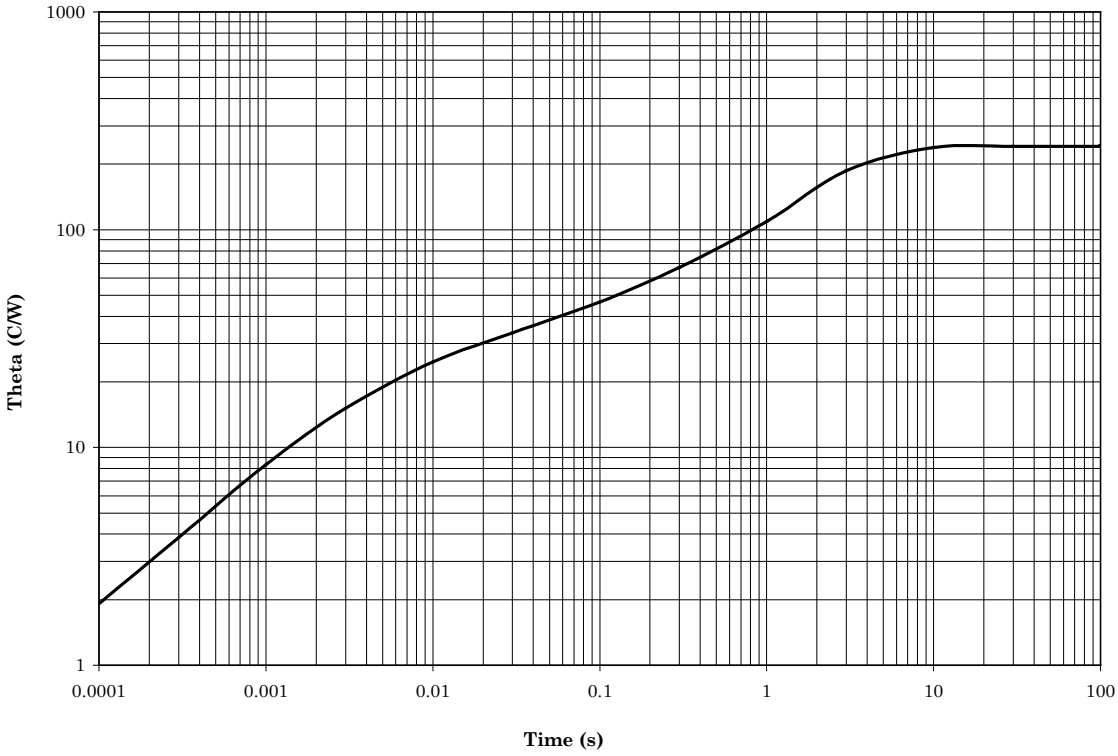
NOTES:

1. The details for the PCB are as follows. Base material is FR4, .0625 inch (1.59 mm) thick, 1-layer 1-oz Cu. PCB pad size for UR devices are .050 inch (1.27 mm) x .087 inch (2.21 mm). PCB land sizes for axial leaded devices are .092 inch (2.34 mm) diameter with L ≤ .125 inch (3.18 mm) lead length. Measurements taken at T_A = 25°C. The pad sizes for UR devices and lead length for axial leaded devices are different than those listed in 1.3 footnote 2.
2. Maximum power rating = 400 mW at T_A =55°C.
3. Thermal resistance junction to ambient (with defined PCB mounting pad) = 300°C/W”

FIGURE 13. Thermal impedance for glass body devices (DO-35 and DO-213AA) when mounted to PCB).

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THERMAL IMPEDANCE
DO-35

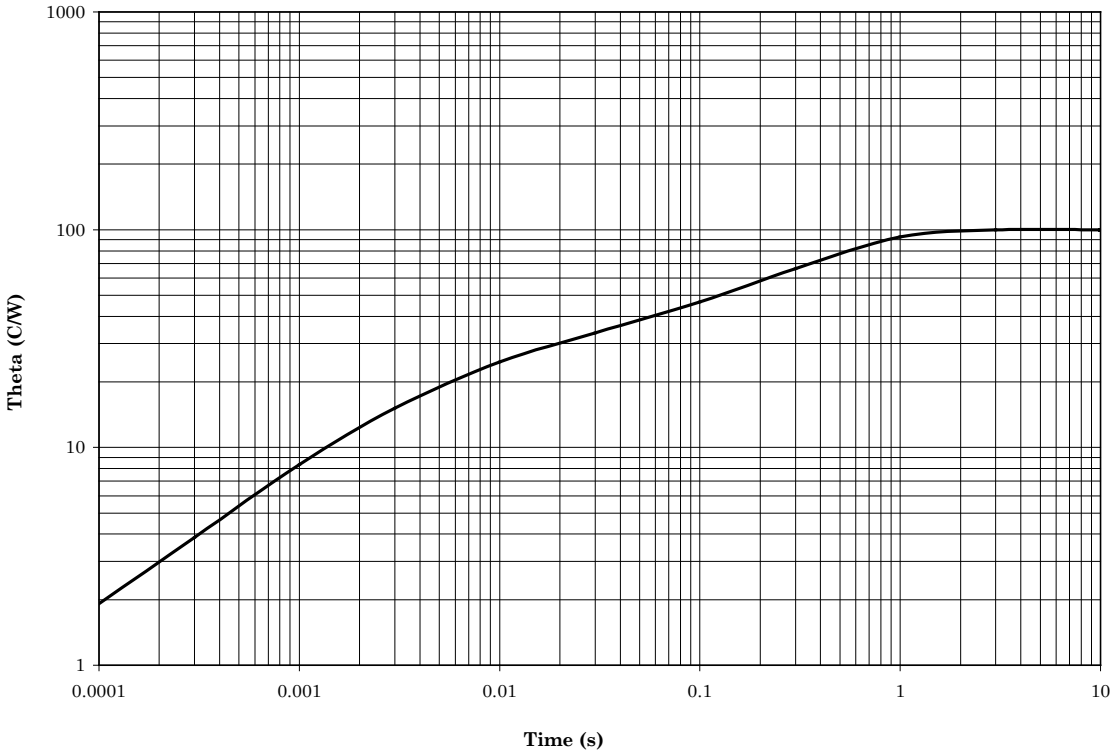


NOTES:
1. T_L = 25°C at .375 inch (9.53 mm) from body.
2. Maximum power rating = 500 mW at T_L = 50°C.
3. Thermal resistance junction-to lead = 250°C/W.

FIGURE 14. Thermal impedance for DO-35 axial leaded device, junction to lead.

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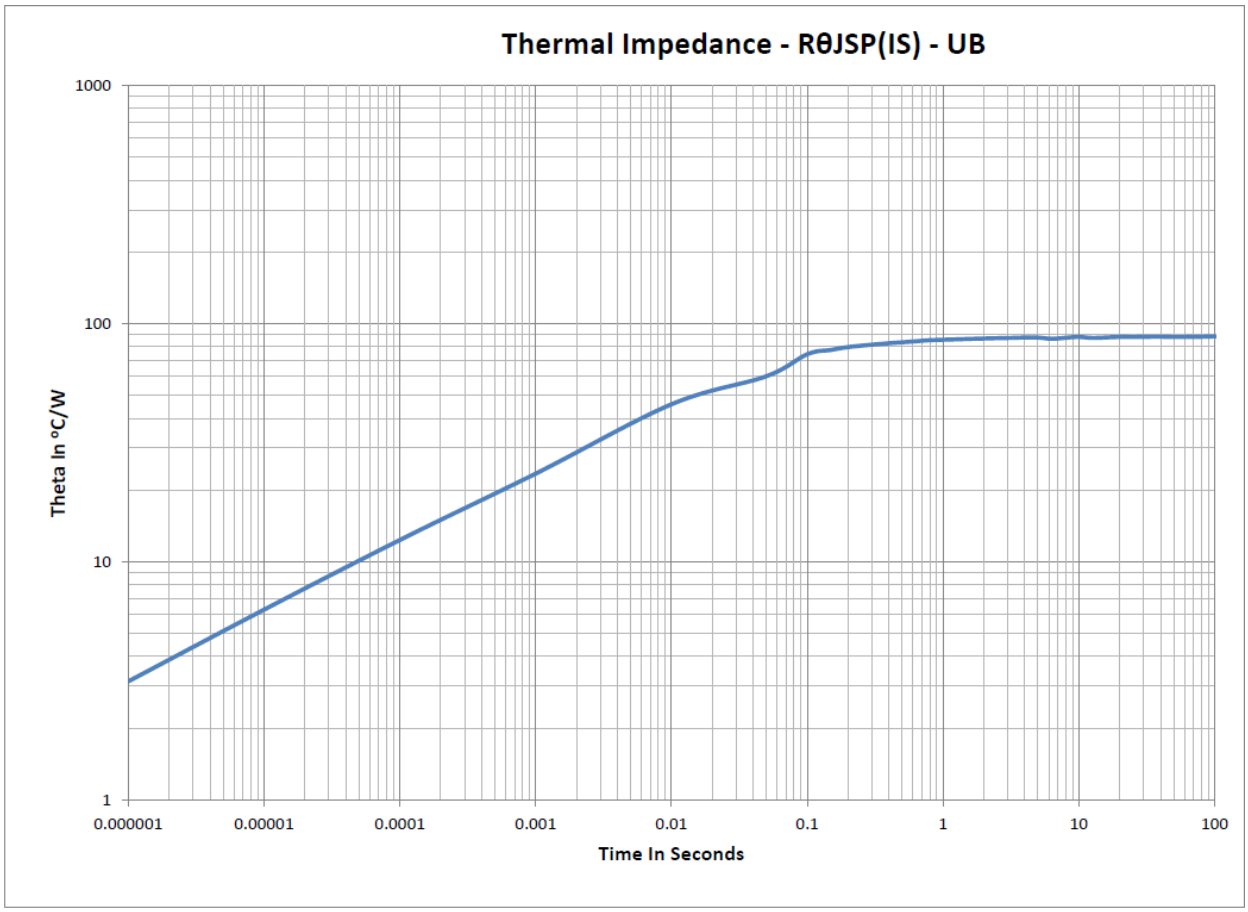
THERMAL IMPEDANCE
DO-213AA



NOTES:
1. Power rating = 500 mW at T_{EC} = 125°C.
2. Thermal resistance junction to end-cap = 100°C/W.

FIGURE 15. Thermal impedance for DO-213AA MELF surface mount device, junction to end-cap.

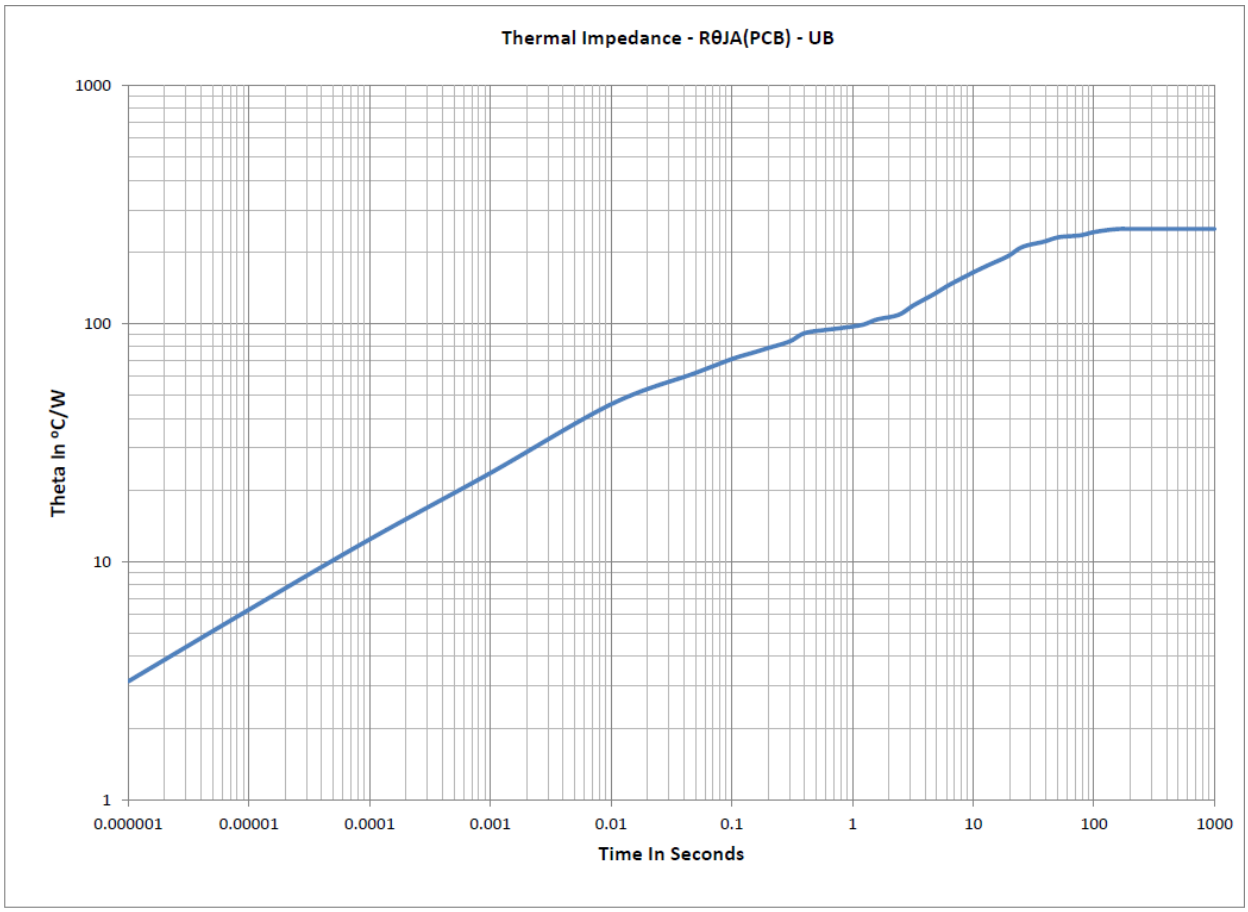
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RθJPCB(IS) = 90°C/W Thermal Impedance
Note: ZθJX = 45 °C/W maximum @10mS

FIGURE 16. Thermal impedance for UB surface mount device, with infinite sink.

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RθJA(PCB) = 250°C/W Thermal Impedance
Note: ZθJX = 45°C/W maximum @10mS

FIGURE 17. Thermal impedance for UB surface mount device, junction to ambient.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

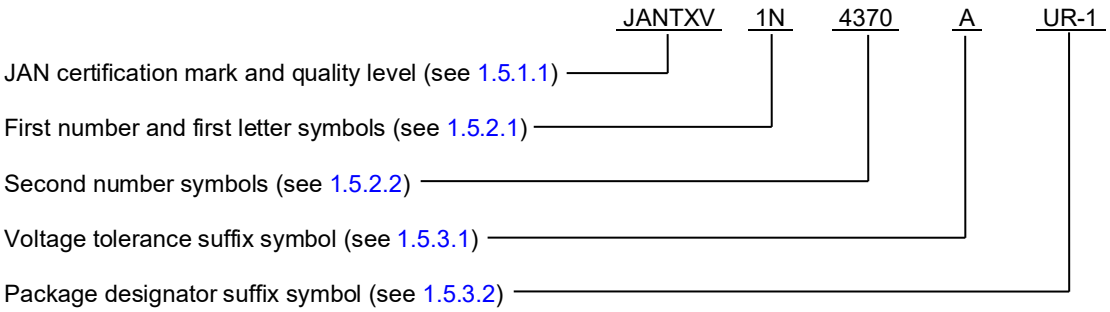
d. Product assurance level and type designator.

e. The complete PIN, see 1.5 and 6.5.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Weapons Support, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

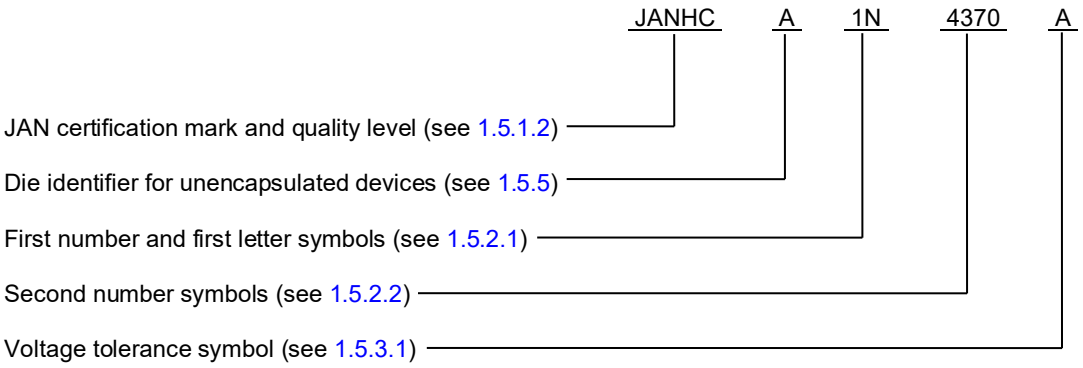
6.4 PIN construction example.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



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6.4.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.5 List of PINs. The following is a list of possible PINs available on this specification sheet.

6.5.1 List of encapsulated device types.

PINs for type 1N4370A-1 through 1N4372A-1 AND 1N746A-1 through 1N759A-1.			
JAN1NA4370A-1	JAN1N746A-1	JANTX1N746A-1	JANTXV1N746A-1
JAN1NA4371A-1	JAN1N747A-1	JANTX1N747A-1	JANTXV1N747A-1
JAN1NA4372A-1	JAN1N748A-1	JANTX1N748A-1	JANTXV1N748A-1
JAN1NA4370C-1	JAN1N749A-1	JANTX1N749A-1	JANTXV1N749A-1
JAN1NA4371C-1	JAN1N750A-1	JANTX1N750A-1	JANTXV1N750A-1
JAN1NA4372C-1	JAN1N751A-1	JANTX1N751A-1	JANTXV1N751A-1
JAN1NA4370D-1	JAN1N752A-1	JANTX1N752A-1	JANTXV1N752A-1
JAN1NA4371D-1	JAN1N753A-1	JANTX1N753A-1	JANTXV1N753A-1
JAN1NA4372D-1	JAN1N754A-1	JANTX1N754A-1	JANTXV1N754A-1
JANTX1NA4370A-1	JAN1N755A-1	JANTX1N755A-1	JANTXV1N755A-1
JANTX1NA4371A-1	JAN1N756A-1	JANTX1N756A-1	JANTXV1N756A-1
JANTX1NA4372A-1	JAN1N757A-1	JANTX1N757A-1	JANTXV1N757A-1
JANTX1NA4370C-1	JAN1N758A-1	JANTX1N758A-1	JANTXV1N758A-1
JANTX1NA4371C-1	JAN1N759A-1	JANTX1N759A-1	JANTXV1N759A-1
JANTX1NA4372C-1	JAN1N746C-1	JANTX1N746C-1	JANTXV1N746C-1
JANTX1NA4370D-1	JAN1N747C-1	JANTX1N747C-1	JANTXV1N747C-1
JANTX1NA4371D-1	JAN1N748C-1	JANTX1N748C-1	JANTXV1N748C-1
JANTX1NA4372D-1	JAN1N749C-1	JANTX1N749C-1	JANTXV1N749C-1
JANTXV1NA4370A-1	JAN1N750C-1	JANTX1N750C-1	JANTXV1N750C-1
JANTXV1NA4371A-1	JAN1N751C-1	JANTX1N751C-1	JANTXV1N751C-1
JANTXV1NA4372A-1	JAN1N752C-1	JANTX1N752C-1	JANTXV1N752C-1
JANTXV1NA4370C-1	JAN1N753C-1	JANTX1N753C-1	JANTXV1N753C-1
JANTXV1NA4371C-1	JAN1N754C-1	JANTX1N754C-1	JANTXV1N754C-1
JANTXV1NA4372C-1	JAN1N755C-1	JANTX1N755C-1	JANTXV1N755C-1
JANTXV1NA4370D-1	JAN1N756C-1	JANTX1N756C-1	JANTXV1N756C-1

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PINs for type 1N4370A-1 through 1N4372A-1 AND 1N746A-1 through 1N759A-1 − continued.			
JANTXV1NA4371D-1	JAN1N757C-1	JANTX1N757C-1	JANTXV1N757C-1
JANTXV1NA4372D-1	JAN1N758C-1	JANTX1N758C-1	JANTXV1N758C-1
	JAN1N759C-1	JANTX1N759C-1	JANTXV1N759C-1
	JAN1N746D-1	JANTX1N746D-1	JANTXV1N746D-1
	JAN1N747D-1	JANTX1N747D-1	JANTXV1N747D-1
	JAN1N748D-1	JANTX1N748D-1	JANTXV1N748D-1
	JAN1N749D-1	JANTX1N749D-1	JANTXV1N749D-1
	JAN1N750D-1	JANTX1N750D-1	JANTXV1N750D-1
	JAN1N751D-1	JANTX1N751D-1	JANTXV1N751D-1
	JAN1N752D-1	JANTX1N752D-1	JANTXV1N752D-1
	JAN1N753D-1	JANTX1N753D-1	JANTXV1N753D-1
	JAN1N754D-1	JANTX1N754D-1	JANTXV1N754D-1
	JAN1N755D-1	JANTX1N755D-1	JANTXV1N755D-1
	JAN1N756D-1	JANTX1N756D-1	JANTXV1N756D-1
	JAN1N757D-1	JANTX1N757D-1	JANTXV1N757D-1
	JAN1N758D-1	JANTX1N758D-1	JANTXV1N758D-1
	JAN1N759D-1	JANTX1N759D-1	JANTXV1N759D-1

PINs for type 1N4370AUR-1 through 1N4372AUR-1 AND 1N746AUR-1 through 1N759AUR-1			
JAN1NA4370AUR-1	JAN1N746AUR-1	JANTX1N746AUR-1	JANTXV1N746AUR-1
JAN1NA4371AUR-1	JAN1N747AUR-1	JANTX1N747AUR-1	JANTXV1N747AUR-1
JAN1NA4372AUR-1	JAN1N748AUR-1	JANTX1N748AUR-1	JANTXV1N748AUR-1
JAN1NA4370CUR-1	JAN1N749AUR-1	JANTX1N749AUR-1	JANTXV1N749AUR-1
JAN1NA4371CUR-1	JAN1N750AUR-1	JANTX1N750AUR-1	JANTXV1N750AUR-1
JAN1NA4372CUR-1	JAN1N751AUR-1	JANTX1N751AUR-1	JANTXV1N751AUR-1
JAN1NA4370DUR-1	JAN1N752AUR-1	JANTX1N752AUR-1	JANTXV1N752AUR-1
JAN1NA4371DUR-1	JAN1N753AUR-1	JANTX1N753AUR-1	JANTXV1N753AUR-1
JAN1NA4372DUR-1	JAN1N754AUR-1	JANTX1N754AUR-1	JANTXV1N754AUR-1
JANTX1NA4370AUR-1	JAN1N755AUR-1	JANTX1N755AUR-1	JANTXV1N755AUR-1
JANTX1NA4371AUR-1	JAN1N756AUR-1	JANTX1N756AUR-1	JANTXV1N756AUR-1
JANTX1NA4372AUR-1	JAN1N757AUR-1	JANTX1N757AUR-1	JANTXV1N757AUR-1
JANTX1NA4370CUR-1	JAN1N758AUR-1	JANTX1N758AUR-1	JANTXV1N758AUR-1
JANTX1NA4371CUR-1	JAN1N759AUR-1	JANTX1N759AUR-1	JANTXV1N759AUR-1
JANTX1NA4372CUR-1	JAN1N746CUR-1	JANTX1N746CUR-1	JANTXV1N746CUR-1
JANTX1NA4370DUR-1	JAN1N747CUR-1	JANTX1N747CUR-1	JANTXV1N747CUR-1
JANTX1NA4371DUR-1	JAN1N748CUR-1	JANTX1N748CUR-1	JANTXV1N748CUR-1
JANTX1NA4372DUR-1	JAN1N749CUR-1	JANTX1N749CUR-1	JANTXV1N749CUR-1
JANTXV1NA4370AUR-1	JAN1N750CUR-1	JANTX1N750CUR-1	JANTXV1N750CUR-1
JANTXV1NA4371AUR-1	JAN1N751CUR-1	JANTX1N751CUR-1	JANTXV1N751CUR-1
JANTXV1NA4372AUR-1	JAN1N752CUR-1	JANTX1N752CUR-1	JANTXV1N752CUR-1
JANTXV1NA4370CUR-1	JAN1N753CUR-1	JANTX1N753CUR-1	JANTXV1N753CUR-1
JANTXV1NA4371CUR-1	JAN1N754CUR-1	JANTX1N754CUR-1	JANTXV1N754CUR-1
JANTXV1NA4372CUR-1	JAN1N755CUR-1	JANTX1N755CUR-1	JANTXV1N755CUR-1
JANTXV1NA4370DUR-1	JAN1N756CUR-1	JANTX1N756CUR-1	JANTXV1N756CUR-1

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PINs for type 1N4370AUR-1 through 1N4372AUR-1 AND 1N746AUR-1 through 1N759AUR-1 – continued.			
JANTXV1NA4371DUR-1	JAN1N757CUR-1	JANTX1N757CUR-1	JANTXV1N757CUR-1
JANTXV1NA4372DUR-1	JAN1N758CUR-1	JANTX1N758CUR-1	JANTXV1N758CUR-1
	JAN1N759CUR-1	JANTX1N759CUR-1	JANTXV1N759CUR-1
	JAN1N746DUR-1	JANTX1N746DUR-1	JANTXV1N746DUR-1
	JAN1N747DUR-1	JANTX1N747DUR-1	JANTXV1N747DUR-1
	JAN1N748DUR-1	JANTX1N748DUR-1	JANTXV1N748DUR-1
	JAN1N749DUR-1	JANTX1N749DUR-1	JANTXV1N749DUR-1
	JAN1N750DUR-1	JANTX1N750DUR-1	JANTXV1N750DUR-1
	JAN1N751DUR-1	JANTX1N751DUR-1	JANTXV1N751DUR-1
	JAN1N752DUR-1	JANTX1N752DUR-1	JANTXV1N752DUR-1
	JAN1N753DUR-1	JANTX1N753DUR-1	JANTXV1N753DUR-1
	JAN1N754DUR-1	JANTX1N754DUR-1	JANTXV1N754DUR-1
	JAN1N755DUR-1	JANTX1N755DUR-1	JANTXV1N755DUR-1
	JAN1N756DUR-1	JANTX1N756DUR-1	JANTXV1N756DUR-1
	JAN1N757DUR-1	JANTX1N757DUR-1	JANTXV1N757DUR-1
	JAN1N758DUR-1	JANTX1N758DUR-1	JANTXV1N758DUR-1
	JAN1N759DUR-1	JANTX1N759DUR-1	JANTXV1N759DUR-1

PINs for type 1N4370AUB through 1N4372AUB AND 1N746AUB through 1N759AUB			
JAN1NA4370AUB	JAN1N746AUB	JANTX1N746AUB	JANTXV1N746AUB
JAN1NA4371AUB	JAN1N747AUB	JANTX1N747AUB	JANTXV1N747AUB
JAN1NA4372AUB	JAN1N748AUB	JANTX1N748AUB	JANTXV1N748AUB
JAN1NA4370CUB	JAN1N749AUB	JANTX1N749AUB	JANTXV1N749AUB
JAN1NA4371CUB	JAN1N750AUB	JANTX1N750AUB	JANTXV1N750AUB
JAN1NA4372CUB	JAN1N751AUB	JANTX1N751AUB	JANTXV1N751AUB
JAN1NA4370DUB	JAN1N752AUB	JANTX1N752AUB	JANTXV1N752AUB
JAN1NA4371DUB	JAN1N753AUB	JANTX1N753AUB	JANTXV1N753AUB
JAN1NA4372DUB	JAN1N754AUB	JANTX1N754AUB	JANTXV1N754AUB
JANTX1NA4370AUB	JAN1N755AUB	JANTX1N755AUB	JANTXV1N755AUB
JANTX1NA4371AUB	JAN1N756AUB	JANTX1N756AUB	JANTXV1N756AUB
JANTX1NA4372AUB	JAN1N757AUB	JANTX1N757AUB	JANTXV1N757AUB
JANTX1NA4370CUB	JAN1N758AUB	JANTX1N758AUB	JANTXV1N758AUB
JANTX1NA4371CUB	JAN1N759AUB	JANTX1N759AUB	JANTXV1N759AUB
JANTX1NA4372CUB	JAN1N746CUB	JANTX1N746CUB	JANTXV1N746CUB
JANTX1NA4370DUB	JAN1N747CUB	JANTX1N747CUB	JANTXV1N747CUB
JANTX1NA4371DUB	JAN1N748CUB	JANTX1N748CUB	JANTXV1N748CUB
JANTX1NA4372DUB	JAN1N749CUB	JANTX1N749CUB	JANTXV1N749CUB
JANTXV1NA4370AUB	JAN1N750CUB	JANTX1N750CUB	JANTXV1N750CUB
JANTXV1NA4371AUB	JAN1N751CUB	JANTX1N751CUB	JANTXV1N751CUB
JANTXV1NA4372AUB	JAN1N752CUB	JANTX1N752CUB	JANTXV1N752CUB
JANTXV1NA4370CUB	JAN1N753CUB	JANTX1N753CUB	JANTXV1N753CUB

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PINs for type 1N4370AUB through 1N4372AUB AND 1N746AUB through 1N759AUB – continued.			
JANTXV1NA4371CUB	JAN1N754CUB	JANTX1N754CUB	JANTXV1N754CUB
JANTXV1NA4372CUB	JAN1N755CUB	JANTX1N755CUB	JANTXV1N755CUB
JANTXV1NA4370DUB	JAN1N756CUB	JANTX1N756CUB	JANTXV1N756CUB
JANTXV1NA4371DUB	JAN1N757CUB	JANTX1N757CUB	JANTXV1N757CUB
JANTXV1NA4372DUB	JAN1N758CUB	JANTX1N758CUB	JANTXV1N758CUB
	JAN1N759CUB	JANTX1N759CUB	JANTXV1N759CUB
	JAN1N746DUB	JANTX1N746DUB	JANTXV1N746DUB
	JAN1N747DUB	JANTX1N747DUB	JANTXV1N747DUB
	JAN1N748DUB	JANTX1N748DUB	JANTXV1N748DUB
	JAN1N749DUB	JANTX1N749DUB	JANTXV1N749DUB
	JAN1N750DUB	JANTX1N750DUB	JANTXV1N750DUB
	JAN1N751DUB	JANTX1N751DUB	JANTXV1N751DUB
	JAN1N752DUB	JANTX1N752DUB	JANTXV1N752DUB
	JAN1N753DUB	JANTX1N753DUB	JANTXV1N753DUB
	JAN1N754DUB	JANTX1N754DUB	JANTXV1N754DUB
	JAN1N755DUB	JANTX1N755DUB	JANTXV1N755DUB
	JAN1N756DUB	JANTX1N756DUB	JANTXV1N756DUB
	JAN1N757DUB	JANTX1N757DUB	JANTXV1N757DUB
	JAN1N758DUB	JANTX1N758DUB	JANTXV1N758DUB
	JAN1N759DUB	JANTX1N759DUB	JANTXV1N759DUB

1/ The pin is also available with the other UB configurations, UBCA, UBD, UBCC, UB2 and UB2R.

6.5.2 List of unencapsulated device types and suppliers of JANHC and JANKC die. The qualified JANHC and JANKC die suppliers with the applicable letter version (example JANHCA1N4370A) will be identified on the QML.

JANHC ordering information (1) (2)					
Type	Manufacture CAGE				
	43611	12954	52GC4	13409	52GC4
1N4370 1N4371 1N4372	JANHCA1N4370A JANHCA1N4371A JANHCA1N4372A	JANHCB1N4370A JANHCB1N4371A JANHCB1N4372A	JANHCC1N4370A JANHCC1N4371A JANHCC1N4372A	JANHCD1N4370A JANHCD1N4371A JANHCD1N4372A	JANHCE1N4370A JANHCE1N4371A JANHCE1N4372A
1N746 1N747 1N748 1N749 1N750 1N751 1N752 1N753 1N754 1N755 1N756 1N757 1N758 1N759	JANHCA1N746A JANHCA1N747A JANHCA1N748A JANHCA1N749A JANHCA1N750A JANHCA1N751A JANHCA1N752A JANHCA1N753A JANHCA1N754A JANHCA1N755A JANHCA1N756A JANHCA1N757A JANHCA1N758A JANHCA1N759A	JANHCB1N746A JANHCB1N747A JANHCB1N748A JANHCB1N749A JANHCB1N750A JANHCB1N751A JANHCB1N752A JANHCB1N753A JANHCB1N754A JANHCB1N755A JANHCB1N756A JANHCB1N757A JANHCB1N758A JANHCB1N759A	JANHCC1N746A JANHCC1N747A JANHCC1N748A JANHCC1N749A JANHCC1N750A JANHCC1N751A JANHCC1N752A JANHCC1N753A JANHCC1N754A JANHCC1N755A JANHCC1N756A JANHCC1N757A JANHCC1N758A JANHCC1N759A	JANHCD1N746A JANHCD1N747A JANHCD1N748A JANHCD1N749A JANHCD1N750A JANHCD1N751A JANHCD1N752A JANHCD1N753A JANHCD1N754A JANHCD1N755A JANHCD1N756A JANHCD1N757A JANHCD1N758A JANHCD1N759A	JANHCE1N746A JANHCE1N747A JANHCE1N748A JANHCE1N749A JANHCE1N750A JANHCE1N751A JANHCE1N752A JANHCE1N753A JANHCE1N754A JANHCE1N755A JANHCE1N756A JANHCE1N757A JANHCE1N758A JANHCE1N759A

(1) Devices are listed with an A suffix only. Suffixes may be ordered with "A", "C", "D", or "E".
(2) Replace "HC" with "KC" when ordering JANKC die.

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6.6 Cross reference substitution.

6.6.1 Quality level JANS devices. Quality level JANS devices, which were covered in revisions prior to MIL-PRF-19500/127K, dated 4 June 1990, will no longer be built to this specification. Quality level JANS devices required for space flight applications are found in MIL-PRF-19500/533. Existing supplies of parts can be used until existing supplies are exhausted. A PIN for PIN replacement table for axial leaded packages follows and these devices are directly interchangeable.

JANS superseded PIN	JANS superseding PIN
1N4370A-1, C-1 or D-1	1N6309, C, D
1N4371A-1, C-1 or D-1	1N6310, C, D
1N4372A-1, C-1 or D-1	1N6311, C, D
1N746A-1, C-1 or D-1	1N6312, C, D
1N747A-1, C-1 or D-1	1N6313, C, D
1N748A-1, C-1 or D-1	1N6314, C, D
1N749A-1, C-1 or D-1	1N6315, C, D
1N750A-1, C-1 or D-1	1N6316, C, D
1N751A-1, C-1 or D-1	1N6317, C, D

JANS superseded PIN	JANS superseding PIN
1N752A-1, C-1 or D-1	1N6318, C, D
1N753A-1, C-1 or D-1	1N6319, C, D
1N754A-1, C-1 or D-1	1N6320, C, D
1N755A-1, C-1 or D-1	1N6321, C, D
1N756A-1, C-1 or D-1	1N6322, C, D
1N757A-1, C-1 or D-1	1N6323, C, D
1N758A-1, C-1 or D-1	1N6324, C, D
1N759A-1, C-1 or D-1	1N6326, C, D

6.6.2 Substitutability of dash-one parts. Non-dash-one devices have been deleted from this specification. Dash-one devices are a direct substitute for non dash-one devices and are preferred. The following table shows the direct substitutability.

Superseding PIN that were specified within MIL-S-19500/127H dated 10 October 1978	Superseded PIN that were specified within MIL-S-19500/127G dated 26 November1975
1N4370A-1	1N4370A
1N4371A-1	1N4371A
1N4372A-1	1N4372A
1N746A-1	1N746A
1N747A-1	1N747A
1N748A-1	1N748A
1N749A-1	1N749A
1N750A-1	1N750A
1N751A-1	1N751A

Superseding PIN that were specified within MIL-S-19500/127H dated 10 October 1978	Superseded PIN that were specified within MIL-S-19500/127G dated 26 November1975
1N752A-1	1N752A
1N753A-1	1N753A
1N754A-1	1N754A
1N755A-1	1N755A
1N756A-1	1N756A
1N757A-1	1N757A
1N758A-1	1N758A
1N759A-1	1N759A

6.6.3 Substitutability of 2 percent and 1 percent tolerance devices. Devices of tighter tolerance are a direct one-way substitute for the looser tolerance devices (example: JANTX1N4370D-1 substitutes for JANTX1N4370A-1).

6.7 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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CONCLUDING MATERIAL

Custodians:
Army − CR
Navy − SH
Air Force − 85
NASA − NA
DLA − CC

Review activities:
Army − AR, AV, MI, SM
Navy − AS, MC
Air Force − 19

Preparing activity:
DLA − CC

(Project 5961−2026−011)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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